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SHORT BIOGRAPHY

Professor of Computer Science and Engineering at King Fahd University of Petroleum and Minerals (KFUPM) in Saudi Arabia, with over 30 years of experience and professional accomplishments in both industry and academia. He is a former fellow of SDAIA-KFUPM Joint Research Center for Artificial Intelligence and Senior Researcher of the Interdisciplinary Research Center on Intelligent Secure Systems. Recognized as a leading expert in Artificial Intelligence, Machine Learning, Cybersecurity, and Networking, his research interests span a wide array of cutting-edge advanced topics, including nature-inspired computing, federated and multimodal deep learning, computer vision, biometrics, data analytics, edge and cloud intelligence, ethical and sustainable intelligence, and various intelligent secure systems applications. Professor El-Alfy's prolific academic career features numerous publications, including books, patents, and papers in peer-reviewed journals and conferences, alongside keynotes, invited talks, and active participation in conference organization, and editorial roles for reputable international publishers such as Artificial Intelligence Review, IEEE/CAA Journal of Automatica Sinica, and IEEE Transactions on Neural Networks and Learning Systems. As a Senior Member of IEEE, an approved ABET/CSAB Program Evaluator, and NCAAA Panel Reviewer, he has made significant contributions to curriculum design and quality assurance in higher education. He is recognized among the top 2% of scientists globally by Stanford University and has received several prestigious awards, including the Excellence in Teaching Award from KFUPM and King Abdullah's Award for Scientific Book Translation. His extensive experience, interdisciplinary research, and high citation record have established him as a highly respected figure and consultant in the field of computer science and engineering.

RESEARCH INTERESTS

- Artificial Intelligence, Machine & Deep Learning
- Edge and Cloud Intelligence, Federated & Multimodal Learning
- Computer Vision & Biometrics
- Cybersecurity & Networking

SELECTED JOURNAL EDITORIAL BOARD

- *Associate Editor*, Artificial Intelligence Review (AIRE), Springer, 2024-2026. [WoS ISI, IF = 10.7, Q1] https://link.springer.com/journal/10462
- *Associate Editor*, International Journal of Machine Learning and Cybernetics, Springer, 2025-2027. [WoS ISI, IF = 2.7, Q2] https://link.springer.com/journal/13042 -- To start in Nov. 2025.
- *Associate Editor*, Artificial Intelligence Science and Engineering (AISE), IEEE xplore. https://mc03.manuscriptcentral.com/aisecn
- *Associate Editor*, Journal of Information and Communication Technology (JICT), 2023-2025. [WoS ISI] https://e-journal.uum.edu.my/index.php/jict/index
- *Associate Editor*, IEEE/CAA Journal of Automatica Sinica; 2018 – 2022. [ISI; Q1] https://www.ieee-jas.net/
- *Associate Editor*, IEEE Transactions on Neural Networks and Learning Systems (previously IEEE Transactions on Neural Networks); 2010 -2016. [ISI; Q1] https://cis.ieee.org/publications/t-neural-networks-and-learning-systems
- *Associate Founding Editor*, International Journal on Trust Management in Computing and Communications, Inderscience, UK. https://www.inderscience.com/jhome.php?jcode=ijtmcc

EDUCATION

- **Ph.D. in Computer Engineering, July 2001** **GPA: 4.0/4.0**
  Stevens Institute of Technology, College of Engineering, Dept. Elect. & Computer Eng., NJ, USA
- **MS in Automatic Control Engineering (specialization in Intelligent Systems), Sept. 1994**
  Mansoura University, College of Eng., Dept. Computer and Control Eng., Mansoura, Egypt
- **BS in Electronics Engineering (with Computer & Control Eng. specialization), May 1991**
  Mansoura University, College of Eng., Dept. Electronics Eng., Mansoura, Egypt

  GPA: **Excellent with honor degree** (Excellent in all the five years)

PUBLICATIONS: PATENTS

1] Salahadin Mohammed, **El-Sayed M. El-Alfy**, Ahmad F. Barradah, Method and System for Querying an XML Database, Application Number: 15/271,848, Filed: Sep 21, 2016, Publication No. 20180081944, Publication Date: Mar 22, 2018, **USPTO GRANT Patent No: 11416547, Grant Date: August 16, 2022**. https://patents.justia.com/patent/11416547

2] **El-Sayed M. El-Alfy** and Sadam Al-Azani, "Multi-modal detection engine of sentiment and demographic characteristics for social media videos", KFUPM Reference #:2018-185, December 26, 2018. Application Number 16/590856, Filed: October 2, 2019. Publication number: 20210103762, Publication Date: April 8, 2021. **USPTO GRANT, Patent number: 11227195, Grant Date: January 18, 2022.** https://patents.justia.com/patent/11227195

3] Luqman, H. A., **El-Alfy, E.-S. M.,** and BinMakhashen, G. "Methods and systems for computerized recognition of hand gestures." Submitted November 30, 2019 KFUPM Ref: 2019-234-TM, Oblon Ref: 527312US. Application Number: 16/809,939, Filed: March 5, 2020, Publication Number: US 20210279453 A1, Publication Date: September 9, 2021. **USPTO GRANT, Patent number: 11227151, Grant Date: January 18, 2022.** https://patents.justia.com/patent/11227151

4] Amer G. A. Binsaadoon and **El-Sayed M. El-Alfy**, Automatic gender recognition utilizing gait energy image (GEI) images", Application Number: 16/393,397, Filed: Apr 24, 2019, Publication Number: 20200342216, Publication Date: October 29, 2020, **USPTO GRANT Patent No. 11138417, Grant Date: October 5, 2021.** https://patents.justia.com/patent/11138417

5] Amer G. A. Binsaadoon and **El-Sayed M. El-Alfy, "**Gait Recognition System to Identify Walking Subject," Application Number: 16/542,351, Filed: Aug 16, 2019, Publication Number: 20190370536, Publication Date: December 5, 2019, **USPTO GRANT, Patent number: 10853633, Date of Patent: December 1, 2020.** https://patents.justia.com/patent/10853633

6] **El-Sayed M. El-Alfy,** and Ali AlHasan, "Spam filtering in multimodal mobile communication," Filed Prov. 62/397564, Sept. 21, 2016. Application Number: 15/711,032, Filed: Sep 21, 2017, Publication No. 20180083903, Publication Date: Mar 22, 2018. Status Date: Jan 18, 2020, Docketed New Case - Ready for Examination. **USPTO GRANT Patent number: 10848448. Date of Patent: November 24, 2020.** https://patents.justia.com/patent/10848448

7] Amer G. A. Binsaadoon and **El-Sayed M. El-Alfy**, "Multi-Kernel Fuzzy Local Gabor Feature Extraction Method for Automatic Gait Recognition," Filed Application No. 15/821121, Nov. 22, 2017, Publication No. 20190156113, Publication Date: May 23, 2019, **USPTO GRANT Patent No. US10552671, Grant Date: February 4, 2020.** https://patents.justia.com/patent/10552671

8] Amer G. A. Binsaadoon and **El-Sayed M. El-Alfy, "**Method for Identifying a Subject Using Gait Analysis," Application Number: 16/542,345, Filed: Aug 16, 2019, Publication Number: US 2019-0370535 A1, Publication Date: December 5, 2019, **USPTO GRANT Patent No. US10540544, Grant Date: January 21, 2020.** https://patents.justia.com/patent/10540544

9] Amer G. A. Binsaadoon and **El-Sayed M. El-Alfy**, "Identification Method Using Partitioning of Subject Silhouettes," Application Number: 16/008,752, Filed: June 14, 2018, Publication Number: 20180293431, Publication Date: October 11, 2018, **USPTO GRANT Patent No. US10140506B2, Grant Date: November 27, 2018.** https://patents.justia.com/patent/20180293431

10] Amer G. A. Binsaadoon and **El-Sayed M. El-Alfy**, "Moving part image partitioning for walking gait identification," Application Number: 16/008,841, Filed: June 14, 2018, Publication Number: 20180293432, Publication Date: October 11, 2018, **USPTO GRANT Patent No. US10115006B1, Grant Date: October 30, 2018.** https://patents.justia.com/patent/10115006

11] Amer G. A. Binsaadoon and **El-Sayed M. El-Alfy**, "Method for identifying an individual by walking style." Application Number: 15/903,866, Application Date: Feb 23, 2018, **USPTO GRANT Patent Number: US10032070B1, Grant Date: July 24, 2018.** https://patents.justia.com/patent/10032070

12] Amer G. A. Binsaadoon and **El-Sayed M. El-Alfy**, "Apparatus and Method for Gait Recognition," U.S. Prov. Application No. 62/276,563, filed on Jan. 8, 2016, Application Number: 15/269,555, Filed: Sep 19, 2016, Publication number: 20180082113, Publication Date: Mar 22, 2018. **USPTO GRANT, Patent Number: US9984284B2**, **Grant Date: 29 May 2018.** https://patents.google.com/patent/US9984284B2/en

13] **El-Sayed M. El-Alfy**, Ahmed Amro, Method and Device for Data Encryption, USPTO Patent, Application Number: 15/241,333, Filed: Aug 19, 2016, Publication No. 20180054301, Publication Date: Feb 22, 2018. https://patents.justia.com/patent/20180054301

PUBLICATIONS: SELECTED JOURNAL PAPERS (OUT OF 80+)

1][Mohammed Ayub, **El-Sayed M. El-Alfy**, "Household Appliance Identification Using Vision Transformers and Multimodal Data Fusion," IEEE Transactions on Consumer Electronics, Vol. 71, No. 2, pp. 2774-2792, May 2025 (First Online: 30 April 2025). **DOI:** 10.1109/TCE.2025.3565850 Available at: https://ieeexplore.ieee.org/document/10980365

2] Sadam Al-Azani, and **El-Sayed M. El-Alfy**, "Integrating Demographic Context in Video Captioning Through Shifted Window Visual Audio Hierarchical Transformer," EAAI, June 2025. [**2nd *Review Round***] [Web of Science, Impact Factor: 7.5 Q1]

3] **El-Sayed M. El-Alfy**, Abdelrahman S. Shbair, Omar A. Al-Oumi, Abdullatif M. Alsaad, Sadam Al-Azani, "A Comparative Study on Face/Body-Based Lie Detection Using Convolutional Neural Networks", Journal of Advances in Information Technology (JAIT), Vol. 16, no. 4, April 2025. Also, presented at 17th International Conference on Computer Science and Information Technology (ICCSIT 2024), Dubai, UAE, October 23-25, 2024. [Web of Science, JIF: 0.9, Q4] doi: 10.12720/jait.16.4.545-555

4] Sadam Al-Azani, **El-Sayed M. El-Alfy**, "A review and critical analysis of multimodal datasets for emotional AI," Artificial Intelligence Review, Vol. 58, Art# 334, pp. 1-65, August 2025. [Web of Science, Impact Factor: 10.7 Q1] https://doi.org/10.1007/s10462-025-11271-1

5] Jianpeng Zang, Jian Wang, Kai Zhang, **El-Sayed M. El-Alfy**, Jacek Mańdziuk, Expertise-informed Bayesian convolutional neural network for oil production forecasting, Geoenergy Science and Engineering, vol. 240, Sept. 2024, 213061. https://doi.org/10.1016/j.geoen.2024.213061 [Web of Science, Impact Factor: NA but it was formerly known as *Journal of Petroleum Science and Engineering*, JIF 4.4 Q3]

6] Xiaoling Gong, Jian Wang, Qilin Ren, Kai Zhang, **El-Sayed M. El-Alfy**, Jacek Ma´ndziuk, Embedded Feature Selection Approach based on TSK Fuzzy System with Sparse Rule Base for High-dimensional Classification Problems, Knowledge-Based Systems, Volume 295, p. 111809, Elsevier, July 2024. https://doi.org/10.1016/j.knosys.2024.111809 [Web of Science Impact Factor: 8.8; Q1]

7] Sadam Al-Azani, Ridhaa Almeshari, **El-Alfy, E.-S. M.,** "Audio-Textual Multi-Label Demographic Recognition of Arabic Speakers Using Deep Learning," *Journal of Intelligent & Fuzzy Systems*, First online April 22nd, 2024. https://doi.org/10.3233/JIFS-219389 [WoS ISI IF = 2.0; Q4]

8] Xuetao Xie, Yi-Fei Pu, Huaqing Zhang, Jacek Mańdziuk, **El-Sayed M. El-Alfy** & Jian Wang, "An Interpretable Neural Network for Robustly Determining the Location and Number of Cluster Centers," International Journal of Machine Learning and Cybernetics, Vol 15, pp. 1473-1501, 2024 (Published first online October 2023). https://doi.org/10.1007/s13042-023-01978-4 [Web of Science IF: 5.6; Q2]

9] Satheesh Abimannan, **El-Sayed M. El-Alfy**, Yue-Shan Chang, Shahid Hussain, Saurabh Shukla, Dhivyadharsini Satheesh, "Ensemble Multifeatured Deep Learning Models and Applications: A Survey," *IEEE Access*, Vol. 11, Sept. 2023, pp. 107194 - 107217. [Web of Science Impact Factor: 3.745; IF5: 4.076; Q2] https://doi.org/10.1109/ACCESS.2023.3320042

10] Satheesh Abimannan, **El-Sayed M. El-Alfy**, Shahid Hussain, Yue-Shan Chang, Saurabh Shukla, Dhivyadharsini Satheesh, and John G. Breslin, "Towards Federated Learning and Multi-Access Edge Computing for Air Quality Monitoring: Literature Review and Assessment," *Sustainability 2023, 15(18), 13951;* https://doi.org/10.3390/su151813951 [WoS ISI IF = 3.9; Q2]

11] Ayub, M., **El-Alfy, E.-S. M.,** "Contextual Sequence-to-Point Deep Learning for Household Energy Disaggregation," *IEEE Access*, Vol. 11, pp. 75599 – 75616, July 2023. [Web of Science Impact Factor: 3.9; Q2] https://doi.org/10.1109/ACCESS.2023.3297552

12] Luqman, H. and **El-Alfy, E.-S. M.,** "Utilizing Motion and Spatial Features for Sign Language Gesture Recognition Using Cascaded CNN and LSTM Models," *Turkish Journal of Electrical Engineering Computer Sciences*, Vol. 30, No. 7, pp. 2508-2525, September 2022. https://doi.org/10.55730/1300-0632.3952

13] **El-Alfy, E.-S. M.,** and Luqman, H., "A Comprehensive Survey and Taxonomy on Sign Language Research," *Engineering Applications of Artificial Intelligence*, Vol. 114, September 2022, 105198. https://doi.org/10.1016/j.engappai.2022.105198

14] Luqman, H., and **El-Alfy, E.-S. M.,** "Towards Hybrid Multimodal Manual and Non-manual Arabic Sign Language Recognition: mArSL Database and Pilot Study", *Electronics*, 10(14), 1739, pp. 1-16, July 2021; https://doi.org/10.3390/electronics10141739. [Web of Science Impact Factor: ; IF5:; Q2]

15] Shah, A., **El-Alfy, E.-S. M.** "Comparative Analysis of Feature Extraction and Fusion for Blind Image Authentication Using Chroma Channels," *Signal Processing: Image Communication*, vol. 95, no. 116271, pp. 1-14, July 2021. https://doi.org/10.1016/j.image.2021.116271

16] Althelaya, K., El-Alfy, E.-S. M., and Mohammed, S., "Combining Deep Learning and Multiresolution Analysis for Stock Market Forecasting," *IEEE Access*, Vol 9, pp. 13099 - 13111, January 2021. https://doi.org /10.1109/ACCESS.2021.3051872 [Web of Science Impact Factor: 3.745; IF5: 4.076; Q1]

17] Luqman, H., **El-Alfy, E.-S. M.,** and BinMakhashen, G. Joint Space Representation and Recognition of Sign Language Fingerspelling Using Gabor Filter and Convolutional Neural Network, Multimedia Tools and Applications, vol. 80, no. 7, March 2021, pp. 10213-10234 (First online: 18 Nov. 2020) https://doi.org/10.1007/s11042-020-09994-0 [WoS IF: 2.313; IF5: 1.994; Q2: CS/SWE]

18] Al-Azani, S., **El-Alfy, E.-S. M.** "Enhanced video analytics for sentiment analysis based on fusing textual, auditory and visual information" IEEE Access, Vol 8, pp. 136843 - 136857, July 2020. https://doi.org/10.1109/ACCESS.2020.3011977 [Web of Science Impact Factor: 3.745; IF5: 4.076; Q1]

19] **El-Alfy, E.-S. M.,** Salahadin Mohammed, S. A review of Machine learning techniques for big data analytics: Bibliometric approach. Technology Analysis & Strategic Management, Published online Feb. 2020, Vol. 32, No. 8, pp. 984-1005, DOI: https://doi.org/10.1080/09537325.2020.1732912 [Web of Science Impact Factor: 1.867; IF5: 2.105; Q3]

20] **El-Alfy, E.-S. M.**, Al-Azani, S., "Empirical Study on Imbalanced Learning of Arabic Sentiment Polarity with Neural Word Embedding," *Journal of Intelligent & Fuzzy Systems,* vol. 38, no. 5, pp. 6211-6222, May 2020. DOI: https://doi.org/10.3233/JIFS-179703 [WoS IF: 1.851; IF5: 1.797; Q3 CS/AI (ESI/CS: Q1)].

21] Al-Obeidat, F. and **El-Alfy, E.-S. M.**, "Hybrid Multicriteria Fuzzy Classification of Network Traffic Patterns, Anomalies, and Protocols," *Personal and Ubiquitous Computing*, First Online: 16 November 2017, vol. 23, no. 5–6, pp 777–791, November 2019. https://doi.org/10.1007/s00779-017-1096-z [Web of Science Impact Factor: 2.0; IF5: 2.212; Q3 CS/IS]

22] Qureshi, M. A. **El-Alfy, E.-S. M.,** "Bibliography of digital image anti-forensics and anti-anti-forensics techniques," IET Image Processing, vol. 13, no. 11, Sept 2019, p. 1811 – 1823. (First Online: 17 June 2019). http://dx.doi.org/10.1049/iet-ipr.2018.6587 [WoS IF: 1.995; IF5: 1.943; Q3 CS; ESI: Q2]

23] **El-Alfy, E.-S. M.**, and Binsaadoon, A. G. "Automated Gait-Based Gender Identification Using Fuzzy Local Binary Patterns with Tuned Parameters," *Journal of Ambient Intelligence and Humanized Computing*, Springer, vol. 10, no. 7, pp. 2495-2504, July 2019. (First Online, 26 Feb. 2018) http://dx.doi.org/10.1007/s12652-018-0728-0 [Web of Science Impact Factor: 4.594; IF5: 3.348; Q1 CS/AI]

24] Naser, H., Awad, W.S., and **El-Alfy, E.-S. M.**, "A Multi-Matching Approximation Algorithm for Symmetric Traveling Salesman Problem," *Journal of Intelligent & Fuzzy Systems,* vol. 36, no. 3, pp. 2285-2295, March 2019. http://dx.doi.org/10.3233/JIFS169939 [WoS IF: 1.851; IF5: 1.782; Q3 CS/AI (ESI/CS: Q1)]

25] **El-Alfy, E.-S. M.,** Baig, Z. A., Abdel-Aal, R. E., "A Novel Approach for Face Recognition Using Fused GMDH-Based Networks," *International Arab Journal of Information Technology* (IAJIT), Vol. 15, No. 3, May 2018, pp. 369 - 377. [*First Online: Apr. 2017*] [WoS IF: 0.467; IF5: 0.654; Q4 CS/AI/IS; Q3 ESI/CS] http://iajit.org/PDF/May%202018,%20No.%203/10828.pdf

26] Al-Utaibi, K. A. and **El-Alfy, E.-S. M.**, "Intrusion Detection Taxonomy and Data Preprocessing Mechanisms," *Journal of Intelligent & Fuzzy Systems*, vol. 34, no. 3, pp. 1369-1383, March 2018 http://dx.doi.org/10.3233/JIFS-169432 [WoS IF: 1.851; IF5: 1.782; Q3 CS/AI (ESI/CS: Q1)]

27] Ahmed, K., **El-Alfy, E.-S. M.**, Awad, W. S. "Ant Colony Inspired Method for Reducing Load Imbalance in Multiprocessor Systems," *Journal of Intelligent & Fuzzy Systems*, vol. 34, no. 3, pp. 1443-1451, March 2018. http://dx.doi.org/10.3233/JIFS-169440 [WoS IF: 1.637; IF5: 1.782; Q3 CS/AI (ESI/CS: Q1)]

28] **El-Alfy, E.-S. M.** "Detection of Phishing Websites Based on Probabilistic Neural Networks and K-Medoids Clustering," *Computer Journal*, Section D: Security in Computer Systems and Networks, Oxford, Vol. 60, No. 12, December 2017, pp. 1745–1759, https://doi.org/10.1093/comjnl/bxx035 [WoS IF: 0.980; IF5: 0.915; Q3/Q4: CS/SWE; ESI/CS: Q1]

29] Baig, M. M., Awais, M. M., **El-Alfy, E.-S. M.** "AdaBoost Based Artificial Neural Network Learning," *Neurocomputing*, Elsevier, Volume 248, July 2017, Pages 120–126. (Published First Online, March 2017). http://dx.doi.org/10.1016/j.neucom.2017.02.077 [WoS IF: 4.438; IF5: 4.01; Q1 CS/AI]

30] **El-Alfy, E.-S. M.** "Evaluation of Sequential Adaptive Testing with Real-Data Simulation: A Case Study," *Journal of Intelligent & Fuzzy Systems*, Vol. 32, No. 4, April 2017, pp. 2977-2986. DOI:10.3233/JIFS-169241 [Web of Science Impact Factor: 1.851; IF5: 1.782; Q3 CS/AI (ESI/CS: Q1)]

31] Baig, M.M., Awais, M.M., **El-Alfy, E.-S. M.** "A Multiclass Cascade of Artificial Neural Network for Network Intrusion Detection," *Journal of Intelligent & Fuzzy Systems*, Vol. 32, No. 4, April 2017, pp. 2875-2883. DOI:10.3233/JIFS-169230 [WoS IF: 1.851; IF5: 1.782; Q3 CS/AI (ESI/CS: Q1)]

32] **El-Alfy, E.-S. M.,** Qureshi, M. A. "Robust Content Authentication of Gray and Color Images Using LBP-DCT Markov-Based Features," *Multimedia Tools and Applications,* Springer, Vol. 76, No. 12, pp. 14535-14556, June 2017 doi:10.1007/s11042-016-3855-7 [WoS IF: 2.313; IF5: 1.994; Q2: CS/SWE]

33] **El-Alfy, E.-S. M.,** Mohammed, S., Barradah, A. F. "XHQE: A Hybrid System for Scalable Selectivity Estimation of XML Queries," *Information Systems Frontiers*, Springer, Vol. 18, No. 6, pp. 1233-1249, Dec. 2016. https://doi.org/10.1007/s10796-015-9561-6 [WoS IF: 3.36; IF5: 4.016; Q2/Q1: CS/IS/TC].

34] **El-Alfy, E.-S. M.,** AlHasan, A. A. "Spam Filtering Framework for Multimodal Mobile Communication Based on Dendritic Cell Algorithm," *Future Generation Computer Systems*, Elsevier, Vol. 64, November 2016, pp. 98-107. http://dx.doi.org/10.1016/j.future.2016.02.018 [WoS IF: 6.125; IF5: 5.387; Q1: CS/TM]

35] **El-Alfy, E.-S. M.**, Alshammari, M. "Towards Scalable Rough Set Based Attribute Subset Selection for Intrusion Detection Using Parallel Genetic Algorithm in MapReduce," *Simulation Modelling Practice and Theory*, Vol. 64, May 2016, pp. 18-29. http://dx.doi.org/10.1016/j.simpat.2016.01.010 [WoS IF: 2.219; IF5: 2.349; Q2/Q1: CS/SWE/INT]

36] Mohammed, S., Barradah, A.F., **El-Alfy, E.-S. M.,** "Selectivity Estimation of Extended XML Query Patterns Based on Prime Number Labeling and Synopsis Modeling," *Simulation Modelling Practice and Theory*, Elsevier, Vol. 64, May 2016, pp. 30-42. http://dx.doi.org/10.1016/j.simpat.2016.01.008 [WoS IF: 2.219; IF5: 2.349; Q2/Q1: CS/SWE/INT]

37] **El-Alfy, E.-S. M.,** Al-Obeidat, F. N. "Detecting Cyber-Attacks on Wireless Mobile Networks Using Multicriterion Fuzzy Classifier with Genetic Attribute Selection," *Mobile Information Systems*, Vol. 2015, 2015. http://dx.doi.org/10.1155/2015/585432 [WoS IF: 1.508; IF5: 1.302; Q4/Q3/Q2/Q1].

38] **El-Alfy, E.-S. M.,** Qureshi, M. A. "Combining Spatial and DCT Based Markov Features for Enhanced Blind Detection of Image Splicing," *Pattern Analysis and Applications*, Vol. 18, No. 3, pp. 713-723, August 2015. doi:10.1007/s10044-014-0396-4 [WoS IF: 1.512; IF5: 1.527; Q3: CS/AI]

39] Mohammed, S., **El-Alfy, E.-S. M.,** Barradah, A. F. "Improved Selectivity Estimator for XML Queries Based on Structural Synopsis," *World Wide Web-Internet and Web Information Systems*, Vol. 18, No. 4, Springer, July 2015, pp. 1123-1144. doi: 10.1007/s11280-014-0311-3 [WoS IF: 2.892; IF5: 2.186; Q2/Q1: CS/SWE/IS].

40] **El-Alfy, E.-S. M.,** Ratrout, N., Gazder, U. "Impact of Stock Market Indices and Other Regional Exogenous Factors on Predictive Modeling of Border Traffic with Neural Network Models," *Arabian Journal for Science and Engineering*, Vol. 40, No. 2, pp 303-312, February 2015. DOI: 10.1007/s13369-014-1438-3 [WoS IF: 1.711; IF5: 1.474; Q3: MULT; Q2: ESI/ENG]

41] **El-Alfy, E.-S. M.,** Abdel-Aal, R. E., Al-Khatib, W. G., Alvi, F., "Boosting Paraphrase Detection through Textual Similarity Metrics with Abductive Networks," *Applied Soft Computing*, Vol. 26, January 2015, pp. 444–453. https://doi.org/10.1016/j.asoc.2014.10.021 [WoS IF: 5.472; IF5: 5.39; Q1: CS/AI].

42] **El-Alfy, E.-S. M.,** Abdel-Aal, R. E., "Abductive Learning Ensembles for Hand Shape Identification," *Cognitive Computation*, Springer, Vol. 6, No. 3, pp. 321-330, September 2014. http://dx.doi.org/10.1007/s12559-013-9241-0 [WoS IF: 4.287; IF5: 3.805; Q1: CS/AI]

43] **El-Alfy, E.-S. M.,** Mujahid, S. N., Selim, S. Z. "A Pareto-based Hybrid Multiobjective Evolutionary Approach for Constrained Multipath Traffic Engineering Optimization in MPLS/GMPLS Networks," *Journal of Network and Computer Applications*, Vol. 36, No. 4, pp. 1196 – 1207, July 2013. http://dx.doi.org/10.1016/j.jnca.2013.02.008 [WoS IF: 5.57; IF5: 5.16; Q1: CS]

44] **El-Alfy, E.-S. M.,** Yao, Y.-D. "Comparing a Class of Dynamic Model-Based Reinforcement Learning Schemes for Handoff Prioritization in Mobile Communication Networks," *Expert Systems with Applications*, Vol. 38, No. 7, July 2011, pp. 8730-8737. http://dx.doi.org/10.1016/j.eswa.2011.01.082 [WoS IF: 5.452; IF5: 5.448; Q1: CS/AI]

45] **El-Alfy, E.-S. M.,** Abdel-Aal, R. E., "Using GMDH-Based Networks for Improved Spam Detection and Email Feature Analysis," *Applied Soft Computing*, Elsevier, Vol. 11, No. 1, January 2011, pp. 477-488. http://dx.doi.org/10.1016/j.asoc.2009.12.007 [WoS IF: 5.472; IF5: 5.39; Q1: CS/AI]

46] Abdel-Aal, R. E., **El-Alfy, E.-S. M.** "Constructing Optimal Educational Tests Using GMDH-Based Item Ranking and Selection," *Neurocomputing*, Vol. 72, 2009, pp. 1184-1197. http://dx.doi.org/10.1016/j.neucom.2008.02.004 [WoS IF: 4.438; IF5: 4.01; Q1 CS/AI]

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PUBLICATIONS: BOOKS (15+ EDITED BOOKS AND TRANSLATED BOOKS)

PUBLICATIONS: CONFERENCE PAPERS (120+)

CITATIONS (TOP 2% STANDFORD LIST)

Pure	https://pure.kfupm.edu.sa/en/persons/elsayed-elalfy
Scopus	https://www.scopus.com/authid/detail.uri?authorId=7003943902
Google Scholar	https://scholar.google.com/citations?user=2a51Yt8AAAAJ&hl=en
RG	ResearchGate: https://www.researchgate.net/profile/El-Sayed-El-Alfy-2
ORCID	https://orcid.org/0000-0001-6279-9776
dblp computer science bibliography	https://dblp.org/pid/71/5614.html
amazon	Books on Amazon: https://www.amazon.com/Books-El-Sayed-M-El-Alfy/s?rh=n%3A283155%2Cp_27%3AEl-Sayed+M.+El-Alfy
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